

Exercise 4

1. Create a confusion matrix from Table 2. Assume that “Y” serves as positive class. Calculate accuracy, precision, recall, F1-measure score, sensitivity and specificity.

Table 2: Actual and predicted labels for a classification problem.

	True Label	Predicted Label
1	N	N
2	N	N
3	N	Y
4	N	N
5	Y	N
6	N	Y
7	N	N
8	Y	Y
9	N	N
10	Y	Y

2. Consider a model that has an accuracy of $p = 85\%$ when evaluated on $N = 200$ test instances. What is the 90% confidence interval for its true accuracy? The formula for the calculation of the confidence interval is given as:

$$p \pm Z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{N}}$$

Refer Table 1 to obtain Z values.

Table 1: Confidence limits for the normal distribution.

$\Pr[X < Z]$	0.70	0.80	0.90	0.95	0.98	0.99
Z	0.52	0.84	1.28	1.64	2.05	2.33

How would the confidence interval change if the size of the test set was 50 or 500, respectively?

- Summarize the performance of the classifiers from Table 3 to calculate the number of wins, draws and losses when comparing the classifiers.

To test whether the difference in performance between a classifier A and a classifier B is significant, consider the following approach. Let N be the size of a given dataset, p_A be the accuracy of classifier A , p_B be the accuracy of classifier B , and

$$p = \frac{p_A + p_B}{2}$$

be the average accuracy for both classifiers. To test whether classifier A is significantly better than B , the following Z -statistic is used:

$$Z = \frac{p_A - p_B}{\sqrt{\frac{2p(1-p)}{N}}}$$

Use a significance level of $\alpha = 5\%$ for the pairwise comparisons.
Hint: Classifier A wins if $Z > 1.64$.

Table 3:

Dataset	N	NB	DT
Congressional Voting Records	435	0.901	0.949
Dermatology	366	0.978	0.956
Employee attrition	1470	0.801	0.805

- An ensemble classifier is a committee of multiple base classifiers. To classify a test instance, the votes from all base classifiers are combined, e.g., using majority voting. Ensemble classifiers frequently outperform their base counterparts in terms of generalization performance. A random forest is an ensemble of decision trees. Explain how a random forest is induced. What are the “random” components of a random forest?
- Explain the following classifier evaluation methods: (a) hold-out method, (b) random subsampling, (c) cross-validation, (d) stratified cross-validation, (e) leave-one-out cross validation and (f) bootstrap.